

Monte Carlo Notes

M08: Lecture 3 (Pages 8-14)

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22 July, 2021

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Monte Carlo methods

- Should view MC methods as numerical integration problems
- Many high dimensional problems can be very hard to integrate numerically, so MC techniques help using stochastic simulation
 - E.g. $E[f(\vec{x}, \vec{\theta})] = \sum_x f(\vec{x}, \vec{\theta}) \Pr(\vec{x}, \vec{\theta}) d\vec{\theta}$ when $\vec{x}$ and/or $\vec{\theta}$ are of the order $10^3 - 10^6$

Classical MC

- We know that because of the strong law of large numbers, the expectation $\frac{1}{n} \sum_{i=1}^n h(X_i) \approx E_f[h(X_1)]$ at large and finite n
- Need to think about Monte Carlo numerical error
 - Use variance of the MC estimator
 - * All the variances are the same, hence we can make that 2nd step (and factor out the $1/n$)
 - * We can estimate the variance term in line 2 from the sample variance $h(X_1), \dots, h(X_n)$ very easily in R
 - * The central limit theorem tells us that error in the Monte Carlo estimate ($\bar{h}_n$) is normally distributed, so we can get the 95% CI using $\bar{h}_n \pm 1.96\sqrt{v_n/n}$

$$\begin{aligned}
 \text{Var}(\bar{h}_n) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n h(X_i)\right) \\
 &= \frac{1}{n^2} \times n \times \text{Var}[h(X_1)] \\
 &\approx \frac{1}{n} \times \frac{1}{n-1} \sum_{i=1}^n [h(X_i) - \bar{h}_n]^2
 \end{aligned}$$

- The issue is that decreasing the MC error scales with $\sqrt{n}$, so to decrease by a factor of 10, we need to increase the number of samples by a factor of 100